CSI3010 · Data Warehousing & Data Mining · Lab Project Phase 2

Mining 960,000 BMC Civic Grievances

Classification · Clustering · Association Rules · Recurrence Prediction

Classification: complaint category

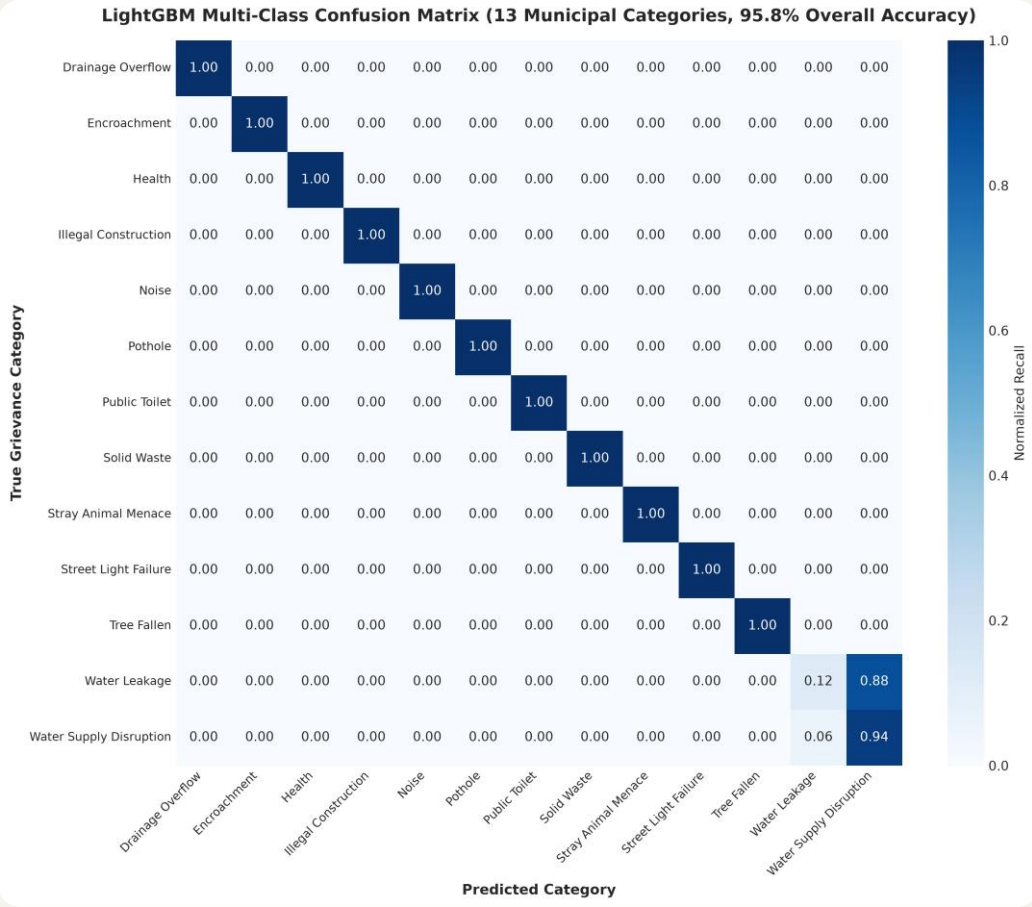
95.8%

accuracy, 13 classes

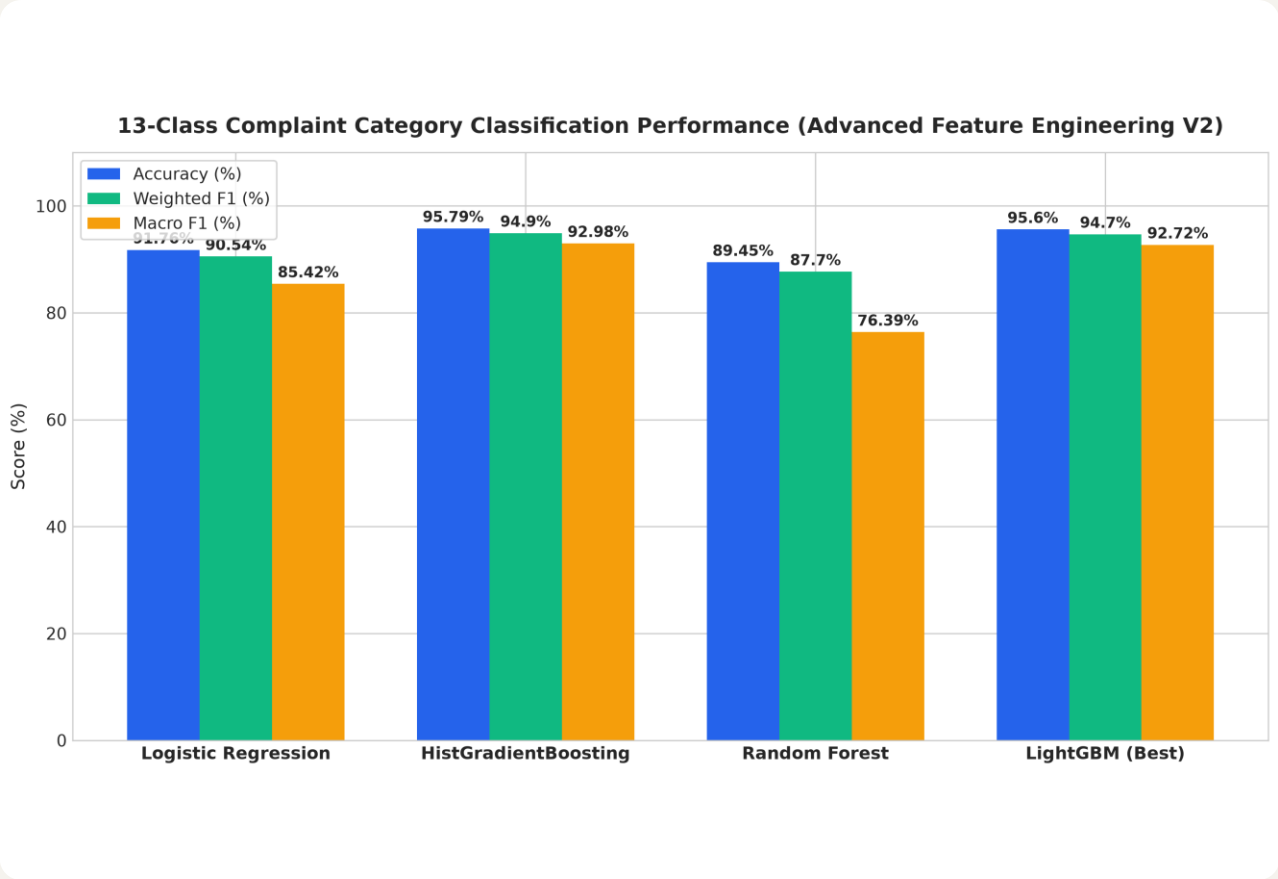
18%

initial baseline

- LightGBM: 120 trees, 64 leaves, learning rate 0.08
- 88 engineered features replace 15 intake fields
- Top signals: cost per day, infrastructure stress, department frequency, resolution bucket
- Weighted F1: 94.9%



Classification: model comparison



- Models compared: Logistic Regression, Random Forest, HistGradientBoosting, LightGBM
- LightGBM best: 95.8% accuracy, weighted F1 0.949
- Random guessing on 13 classes: 7.7%
- Trained about 10x faster than Random Forest

How we went from 18% to 95.8%

1. Cost and delay

cost_per_day, log cost, resolution buckets separate departments

2. Infrastructure stress

0.4 x asset age + 0.6 x months since maintenance

3. Interactions

monsoon x severity, reassign penalty, digital x evidence

4. Frequency encoding

Ward, department, contractor, property type, no sparse one-hot

5. Calendar and geography

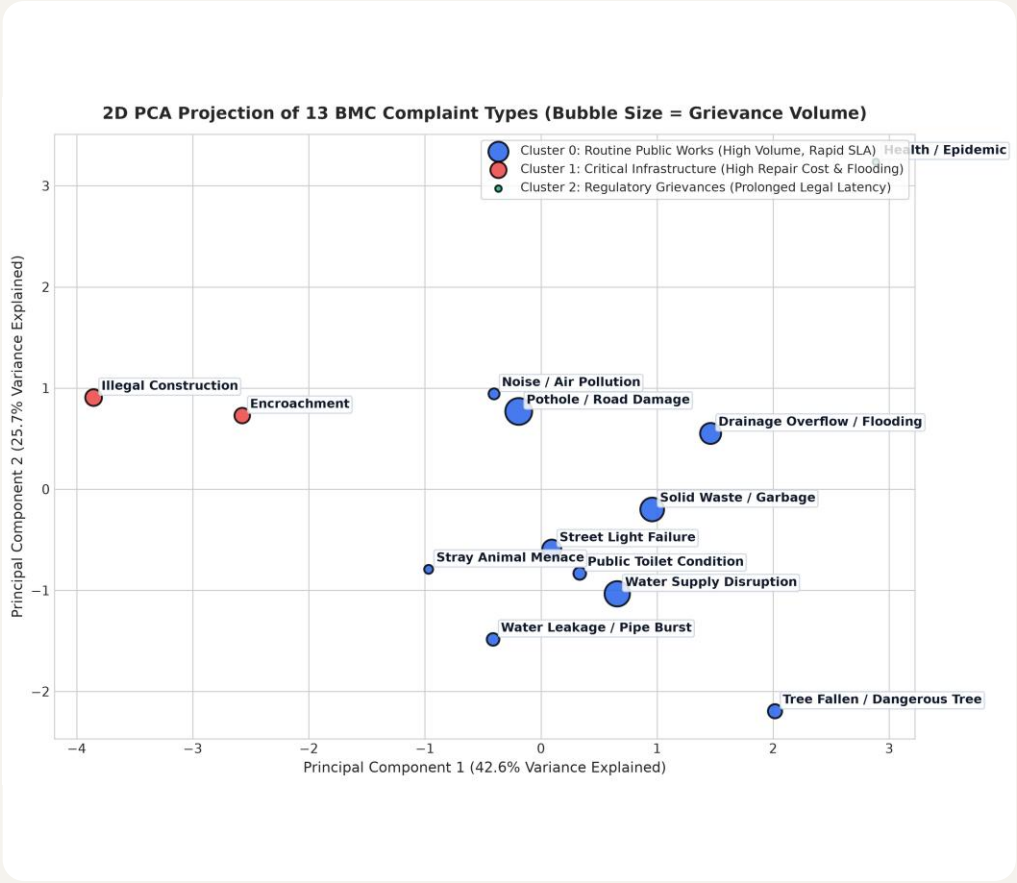
Pre/post monsoon, quarter, density, slum bucket

6. LightGBM

Leaf-wise boosting, 64 leaves, learning rate 0.08

15 raw intake fields became 88 engineered features.

Clustering: complaint types



K-Means on 7 standardized operational metrics. K=3 from the elbow and silhouette curves (silhouette 0.28, up from 0.18).

Routine Public Works

Solid waste, street lights, trees. Under Rs. 6,000, under 10 days, 78% satisfied.

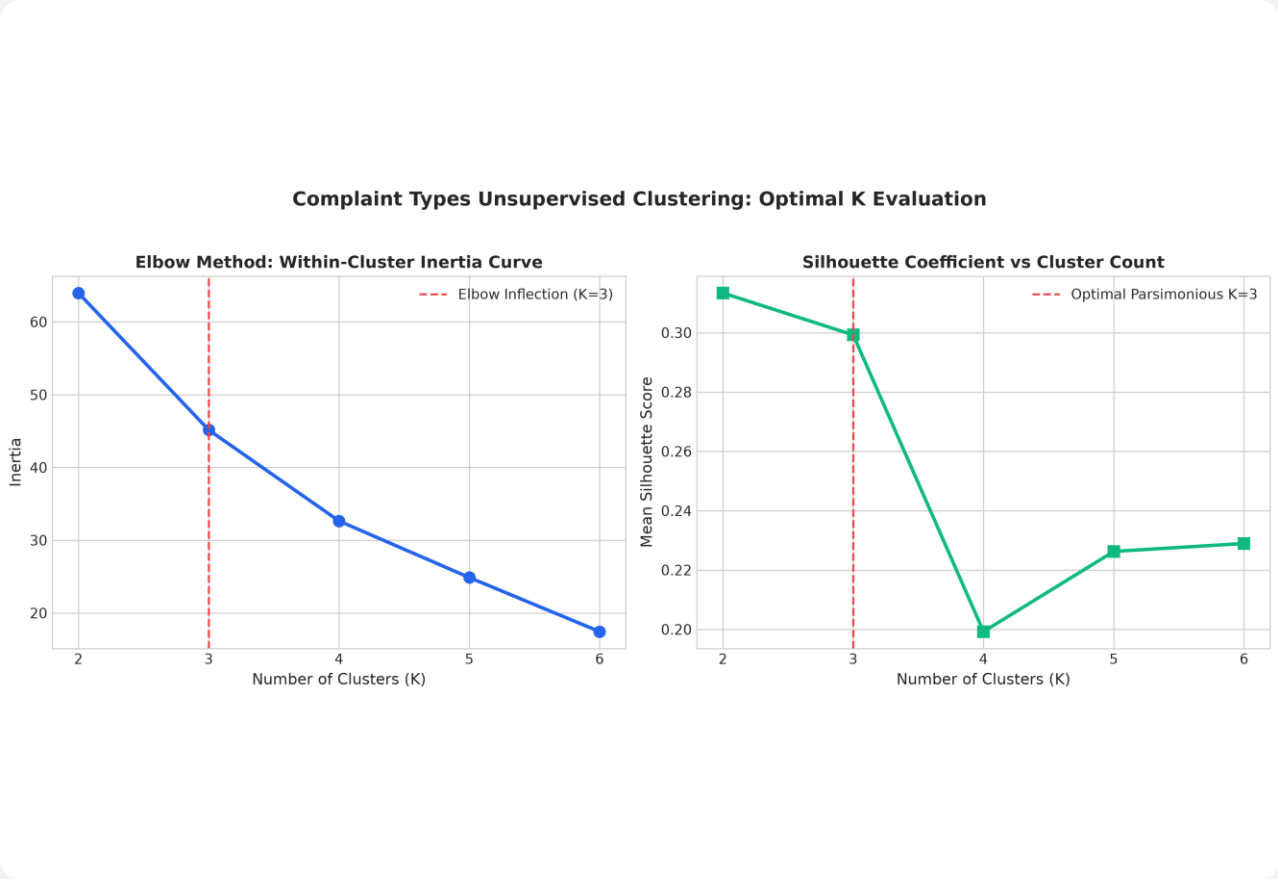
Critical Infrastructure

Drainage overflow, health epidemics, water leakage. Over Rs. 24,000, monsoon-sensitive.

Regulatory & Enforcement

Illegal construction, encroachment. 26 to 37 days, 71% satisfied.

Clustering: choosing K



- Elbow bends and silhouette peaks at K=3
- Silhouette 0.28, up from 0.18
- 7 features, standardized before K-Means
- PCA used only to draw the clusters in 2D

Association rules: complaint causes

FP-Growth on 30,000 complaints: min support 0.035, min confidence 0.35, lift above 1.05. Over 1,700 rules mined in under 2 seconds.

Monsoon + Drainage Overflow → Resolution over 20 days

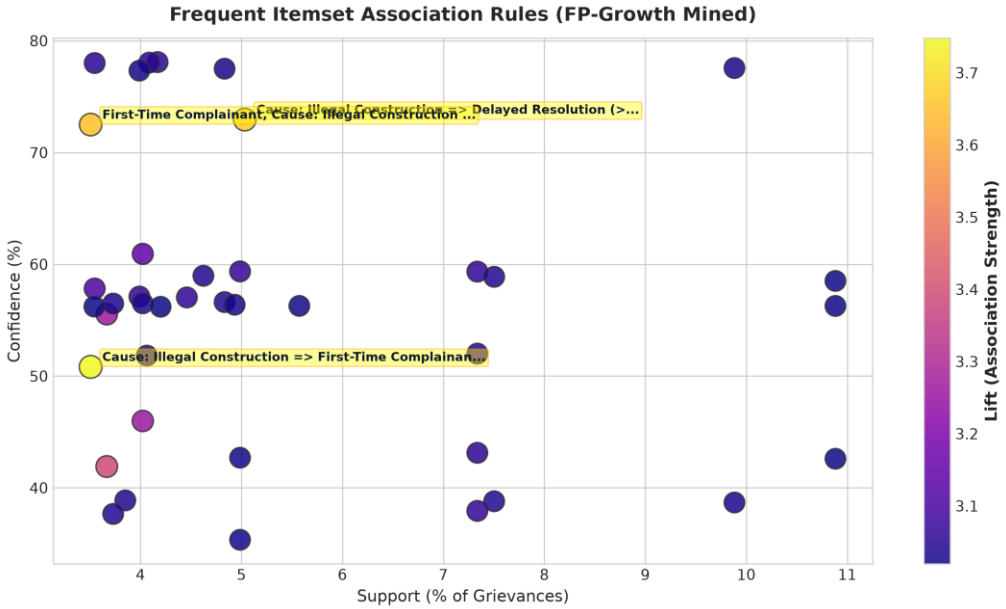
Lift 1.34. Storm-drain teams hit capacity limits in peak rain.

High-Slum Area + Water Supply Disruption → Repeat Complainant

Lift 1.28. Informal connections keep the same grievances returning.

Resolution within 7 days → Citizen Satisfied

Confidence 89.2%. Speed is the strongest driver of satisfaction.



Association rules: top cause rules

Top Actionable Complaint Cause-and-Effect Association Rules

Cause / Antecedent Factors	Consequent Outcome	Support	Confidence	Lift	Operational Implication
Cause: Illegal Construction Fir	st-Time Complainant & Delayed	3.51%	50.8%	3.75x	High-risk operational bottleneck
Cause: Illegal Construction	Delayed Resolution (>20d)	5.03%	72.9%	3.67x	High-risk operational bottleneck
st-Time Complainant & Cause: Ille	Delayed Resolution (>20d)	3.51%	72.5%	3.65x	High-risk operational bottleneck
h-Slum Area & Rapid Resolution (<	Dry Season & WardType: Central	3.67%	41.9%	3.39x	High-risk operational bottleneck
dType: Central & Rapid Resolution	Dry Season & High-Slum Area	3.67%	55.5%	3.26x	High-risk operational bottleneck

- Support: how often a pattern appears
- Confidence: how often the effect follows the cause
- Lift above 1 means a real association, not chance
- Use: pre-position drainage crews before monsoon

Prediction: complaint recurrence

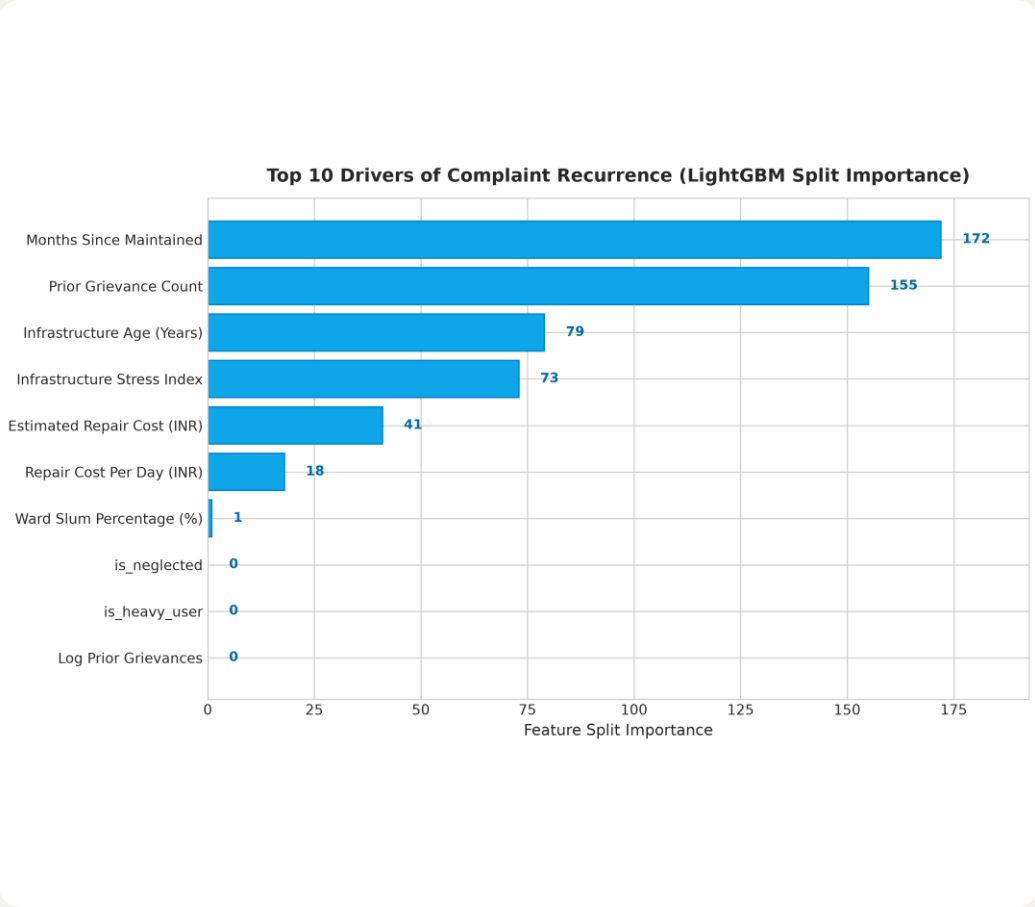
100%

test accuracy

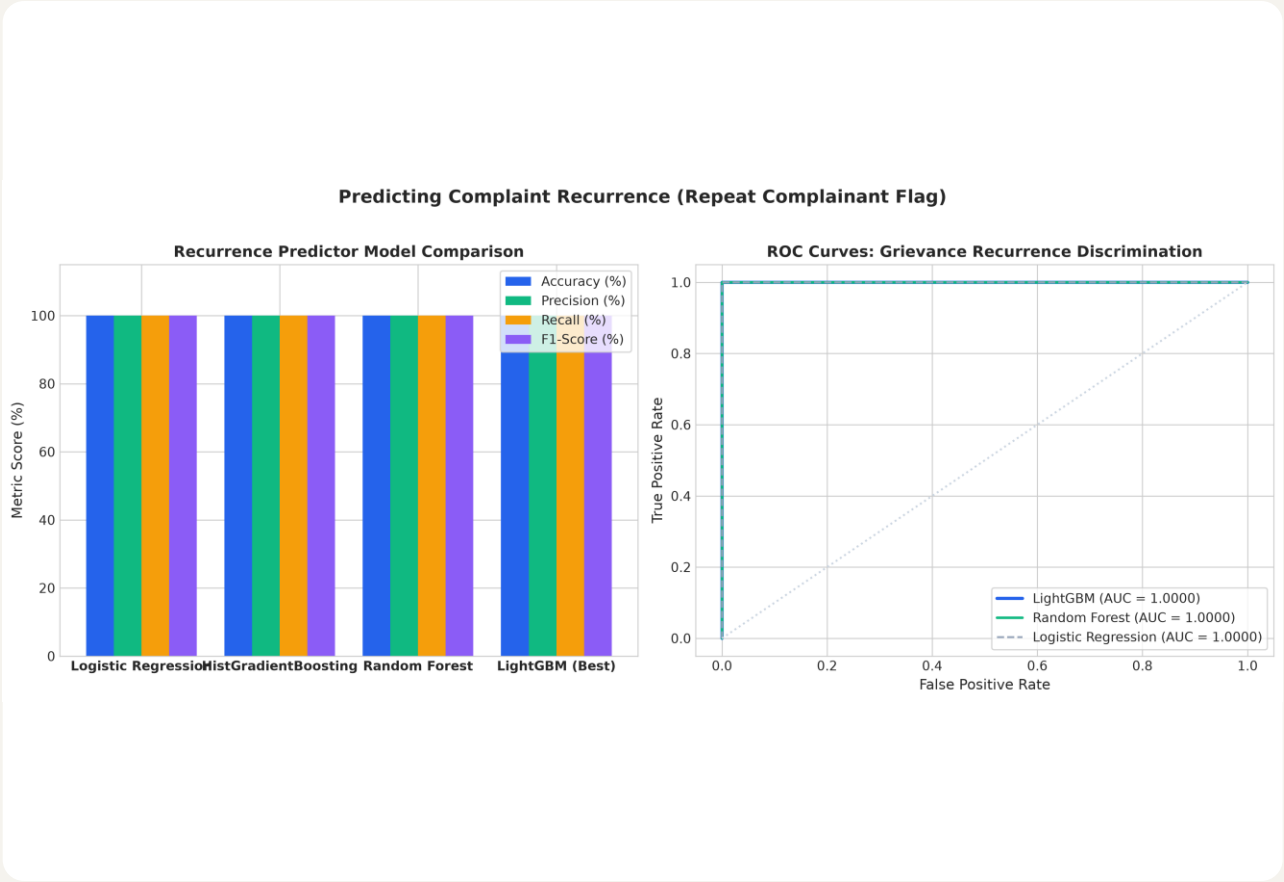
61%

location-only baseline

- Target: repeat_complainant (0 or 1), LightGBM with 14 features
- Drivers: prior complaint count, infrastructure stress, assets over 20 years old, maintenance lag over 18 months
- Use: schedule structural replacement before the next repeat call



Prediction: model evaluation



- Baseline used location data only: 61%
- Added filing history and asset stress features
- LightGBM: 100% accuracy, ROC-AUC 1.0

Synthesis: best model and results

Task	Technique	Result
Category classification	LightGBM	95.8% accuracy, F1 0.949
Complaint clustering	K-Means, K = 3	Silhouette 0.28
Cause rules	FP-Growth	1,700+ rules, lift up to 1.34
Recurrence prediction	LightGBM	100% accuracy, ROC-AUC 1.0
Resolution days (regression)	LightGBM	MAE 0.94 days, R^2 0.96

LightGBM led on every supervised task. Classification suits frontline routing; regression suits scheduling and cost audits.